

Marine Plastic Debris Detection using Machine Learning

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Problem Statement

- Marine plastic pollution threatens ecosystems and human health globally.
- Ship-based surveys are costly, slow, and geographically limited.
- Plastics spectrally resemble sea foam, algae and sun glint in satellite imagery.

Objectives

- To develop a UNet++ deep learning model to automatically detect and segment marine plastic debris in Sentinel-2 satellite imagery.
- To train the model on the MARIDA benchmark dataset using imbalance-aware loss functions with weighted positive class strategies.
- To evaluate models performance using Accuracy, F1 Score, Precision, Recall and IoU metrics.

Methodology

- Acquired Sentinel-2 L2A multispectral satellite imagery and prepared the MARIDA benchmark dataset for training.
- Evaluated results using standard metrics and visualized detections through the DebrisNet web dashboard.

Implementation

- Developed DebrisNet web dashboard using HTML5, CSS3 and JavaScript for interactive marine debris visualization.
- Applied Albumentations library for data augmentation including flips, rotations, brightness contrast shifts and Gaussian noise.
- Performed 128×128 tile-based inference on real Sentinel-2 scenes generating georeferenced GeoTIFF probability maps and binary debris masks.



Fig 1: South Africa costal area

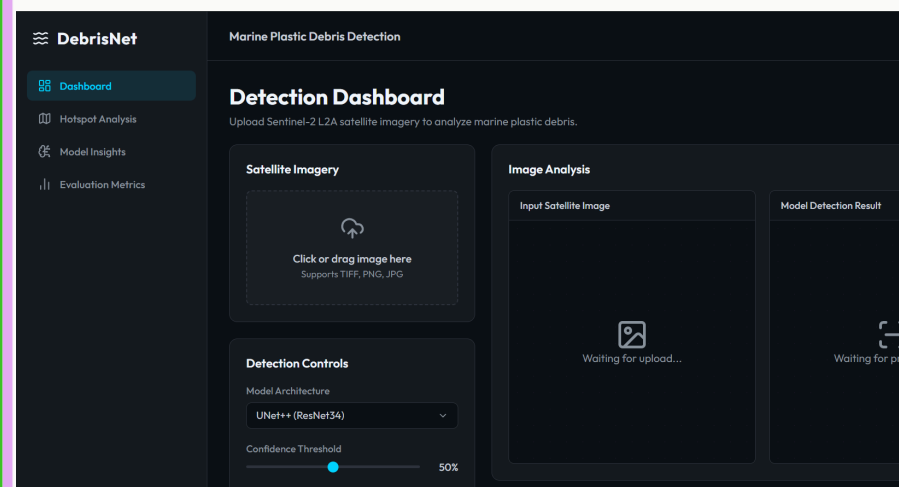


Fig 2:web dashboard

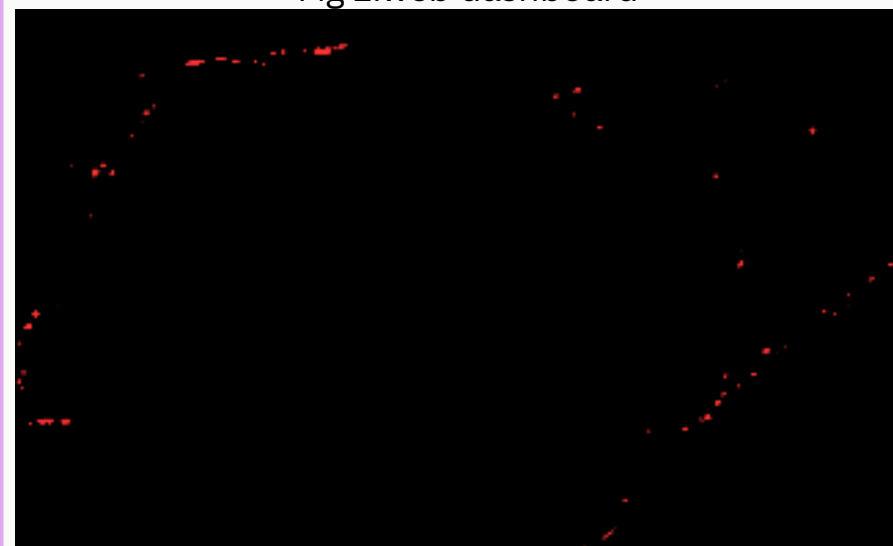


Fig 3:debeis detected

Result

- Model achieved 99.90% accuracy, 14.74% F1 and 7.96% IoU on MARIDA test set.
- Detected 2,712 debris pixels (Motagua) and 56 pixels (Durban) with spatially consistent patterns.
- DebrisNet dashboard successfully visualized detection results with live statistics.

Conclusion

DebrisNet successfully integrates UNet++ segmentation, Sentinel-2 multispectral imagery, and an interactive web dashboard into a unified, scalable pipeline for automated marine plastic debris detection and visualization.

Future Work

- GPU-accelerated training to improve IoU toward 0.30–0.55 benchmarks..
- Live inference backend via Flask/FastAPI for real-time detection

References

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